

Explainable AI

Explain your model and data without coding



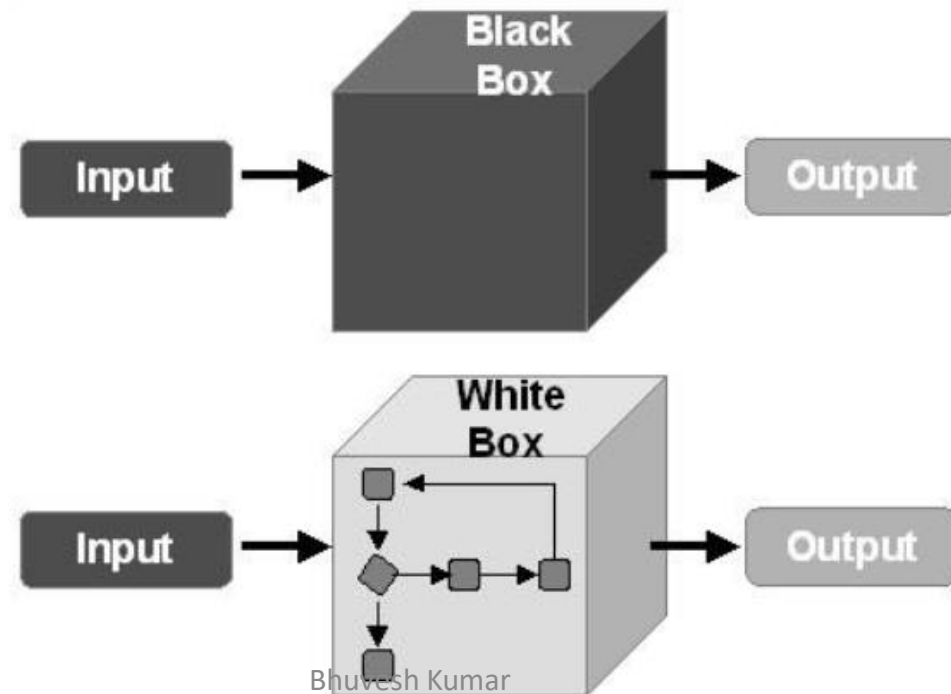
Background

What is an AI model?

- AI/ML models are mathematical algorithms that are “trained” using data

Types of models based on transparency

- White Box model
- Black Box model



Background

Modern Machine Learning models are mostly black-box: only the input features and output prediction are available for humans, the inner working is too complex to be comprehend.



Problem Statement

AI and ML models are being rapidly deployed all around the globe and most of the companies or individuals does not even understand the issues related to unexplained model.

- Lack of research in XAI (Explainable AI) area.**
- Lack of XAI Model-agnostic technique suites.**
- No platform for novice users to help them explain the model.**
- Lack of audience oriented explainable AI models.**
- Lack of trust for AI/ML models (Fintech News, 2019).**

Problem Statement



IBM's Watson for Oncology was criticized by oncologists for delivering unsafe and inaccurate recommendations (Ross and Swetliz [2018](#)).



Amazon's system for curriculum-vitae screening was found to be biased against women (Dastin [2018](#)).



Algorithms behind the Apple Credit Card are accused of being gender-biased (Duffy [2019](#)).



Some tools for sentiment analysis are suspected of age-bias (Diaz et al. [2018](#)).

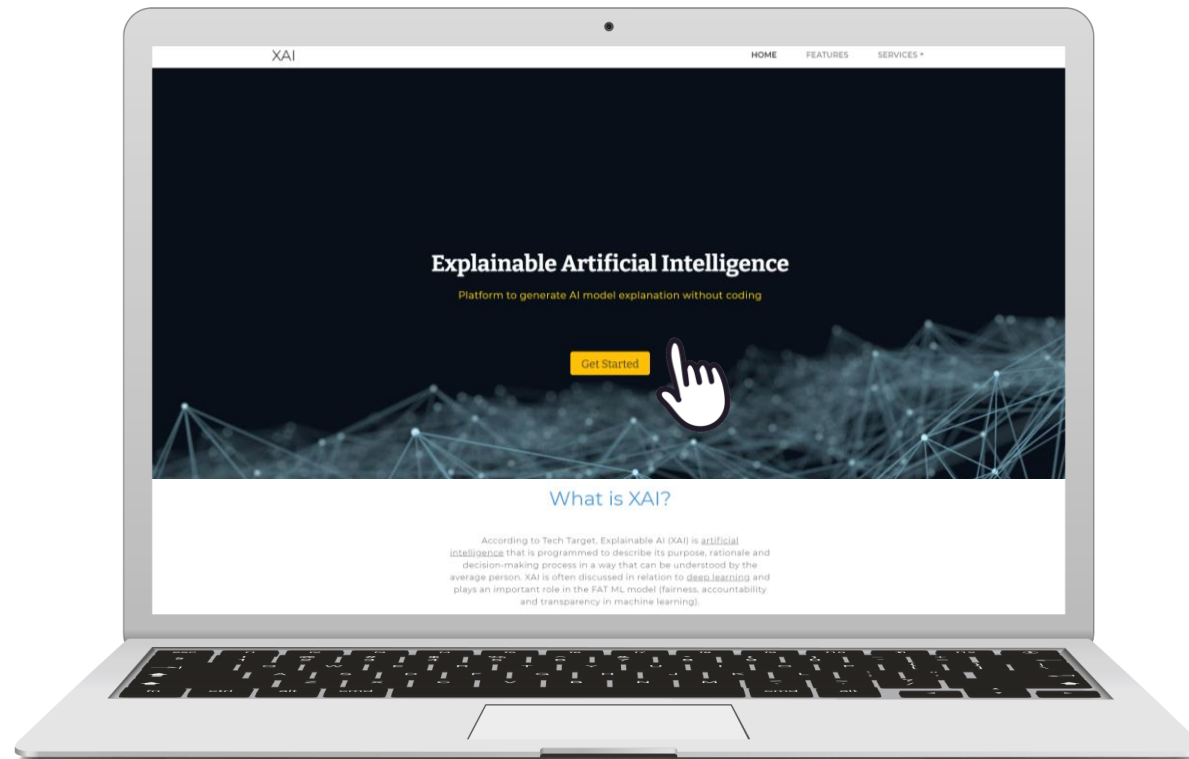
Solution

Web Application

An easy to use web application, where developer can upload the ML/AI model and data and get the explanations (Global and local) by using various State-of-Art techniques like SHAP, LIME etc.

XAI Model-agnostic technique suites.

platform for novice users



audience oriented explanation

Develop trust for AI/ML models

Literature review to add knowledge in XAI domain

Solution

User should locally train the model and store model, X_{test} and y_{test} in pickle files. Then User can easily use the services offered by the web application.



Solution

Literature review

Paper Title: Explainable Artificial Intelligence (XAI): Concepts, taxonomies, opportunities and challenges toward responsible AI (Alejandro Barredo Arrieta, 2020)

Paper Name: Definitions, methods, and applications in interpretable machine learning (W. James Murdoch, 2019)

Paper Name: A Framework for Understanding Unintended Consequences of Machine Learning (Harini Suresh, 2020)

Paper Title: Getting Fairness Right: Towards a Toolbox for Practitioners (Boris Ruf, 2020)

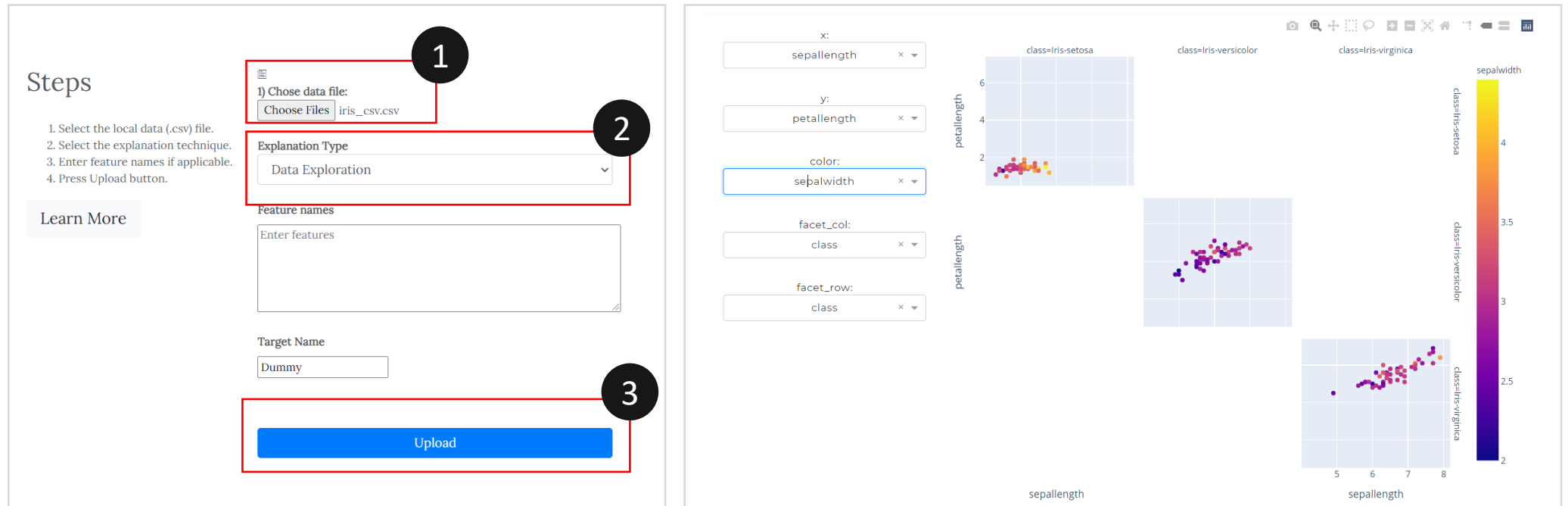
Paper Title: Self-explaining AI as an alternative to interpretable AI (Elton, 2020)

Paper Title: Who is this Explanation for? Human Intelligence and Knowledge Graphs for explainable AI (CELINO, 2020)

01 Service - 1 Data Explanation

Data Exploration

User can upload data file (in .csv format) and explore data in 5 dimensions.
X axis, Y Axis, Z – colour and facet column/row



01 Service - 2 Data Explanation

PCA 2 components

User can upload data file (in .csv format) and add feature names and target name to get PCA 2D plot. (with third dimensions as colour)

Steps

1. Select the local data (.csv) file.
2. Select the explanation technique.
3. Enter feature names if applicable.
4. Press Upload button.

[Learn More](#)

1) Chose data file:
 iris_csv.csv

Explanation Type
PCA 2 Components

Feature names
'sepalength', 'sepalwidth', 'petallength'

Target Name
'class'



01 Service - 3 Data Explanation

PCA 3 components

User can upload data file (in .csv format) and add feature names and target name to get PCA 3D plot. (with another dimensions as colour)

Steps

1. Select the local data (.csv) file.
2. Select the explanation technique.
3. Enter feature names if applicable.
4. Press Upload button.

[Learn More](#)

1

2

3

4

5

1) Chose data file:
 iris_csv.csv

Explanation Type
PCA 3 Components

Feature names
'sepallength', 'sepalwidth', 'petallength'

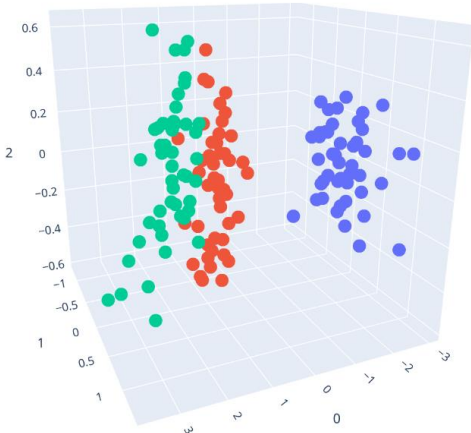
Target Name
'class'

Upload

Select dimension to be shown in colour:

color

- Iris-setosa
- Iris-versicolor
- Iris-virginica



01 Service - 4 Data Explanation

T-SNE

User can upload data file (in .csv format) and add feature names and target name to get T-SNE 3D plot. (with another dimensions as colour)

Steps

1. Select the local data (.csv) file.
2. Select the explanation technique.
3. Enter feature names if applicable.
4. Press Upload button.

[Learn More](#)

1

2

3

4

5

1) Chose data file:

Choose Files iris_csv.csv

Explanation Type

T-SNE

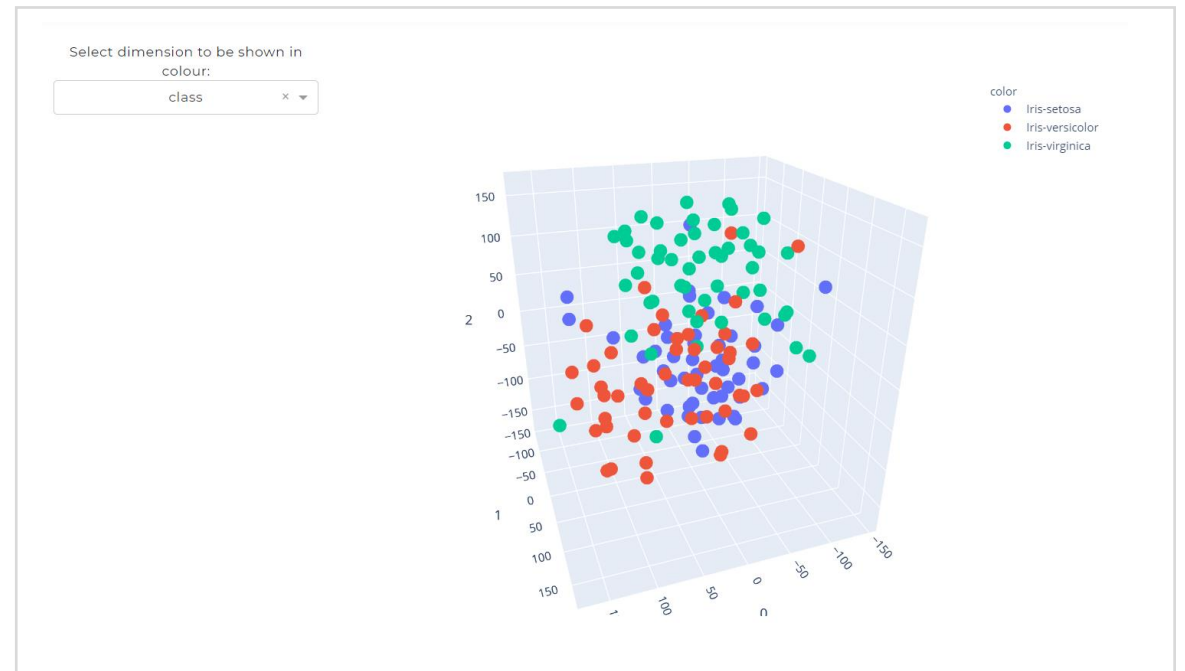
Feature names

'sepalength', 'sepalwidth', 'petallength'

Target Name

'class'

Upload



02 Service - 1 Model Explanation

Global Explanation

User can upload pickle model and data files to get SHAP powered Global Explanations.

Steps

1. Select the local data (.csv) file.
2. Select the explanation technique.
3. Enter feature names if applicable.
4. Press Upload button.

[Learn More](#)

1

1) Chose model file:
Choose Files rf_Model.pkl

1) Chose x test file:
Choose Files X_rf.pkl

1) Chose y test file:
Choose Files y_rf.pkl

2

Explanation Type
Global Explanation

3

Model Type
Random Forrest Regressor

4

Upload

Permutation Importance

Feature importance refers to techniques that assign a **score** to **input features** based on how **useful they are at predicting** a target variable.

[Learn More](#)

Weight	Feature
0.3006 ± 0.0343	sqft_living15
0.2670 ± 0.0218	lat
0.2210 ± 0.0159	sqft_living
0.1350 ± 0.0506	view
0.0511 ± 0.0058	grade
0.0328 ± 0.0070	long
0.0229 ± 0.0045	yr_built
0.0189 ± 0.0028	sqft_above
0.0153 ± 0.0020	zipcode
0.0114 ± 0.0029	sqft_lot
0.0108 ± 0.0026	sqft_basement
0.0100 ± 0.0006	waterfront
0.0097 ± 0.0020	sqft_lot15
0.0085 ± 0.0013	yr_renovated
0.0056 ± 0.0020	condition
0.0048 ± 0.0015	bathrooms
0.0029 ± 0.0009	bedrooms
0.0015 ± 0.0007	floors

02 Service - 2 Model Explanation

Local Explanation

User can upload pickle model and data files to get SHAP powered local Explanations.

Steps

1. Select the local data (.csv) file.
2. Select the explanation technique.
3. Enter feature names if applicable.
4. Press Upload button.

Learn More

1

1) Chose model file:
Choose Files rf_Model.pkl

1) Chose x test file:
Choose Files X_rf.pkl

1) Chose y test file:
Choose Files y_rf.pkl

2

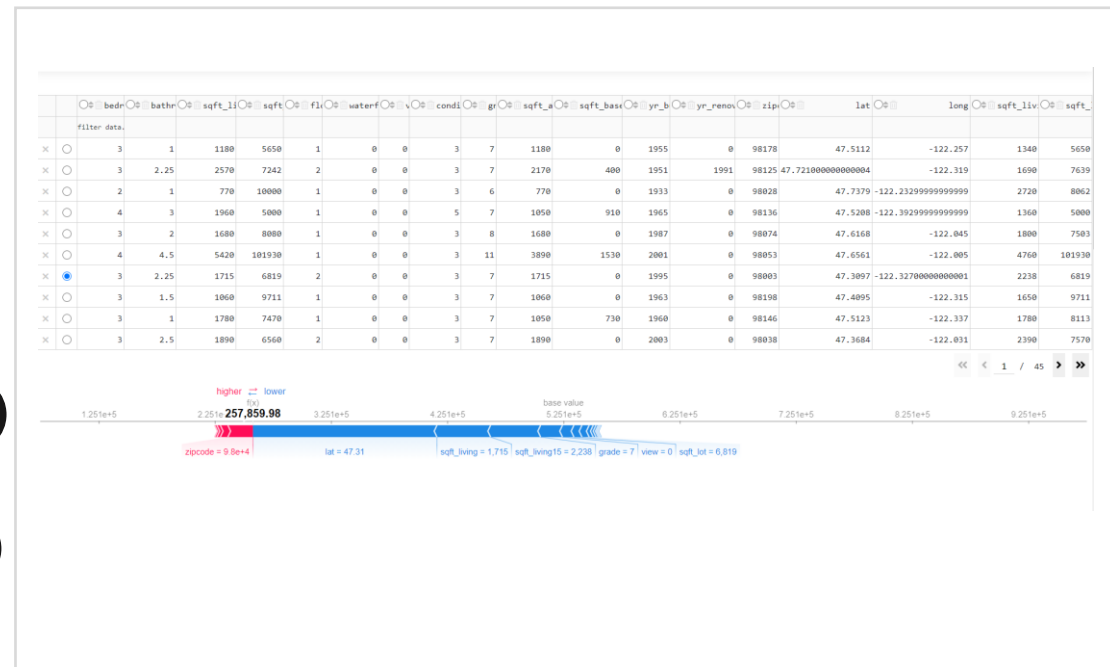
Explanation Type
Local Explanation

3

Model Type
Random Forrestr Regressor

4

Upload



02 Service - 3 Model Explanation

What-if

User can upload pickle model and data files to get SHAP powered local Explanations for user configured instances.

Steps

1. Select the local data (.csv) file.
2. Select the explanation technique.
3. Enter feature names if applicable.
4. Press Upload button.

[Learn More](#)

1

1) Chose model file:
Choose Files rf_Model.pkl

1) Chose x test file:
Choose Files X_rf.pkl

1) Chose y test file:
Choose Files y_rf.pkl

2

Explanation Type
What-if

3

Model Type
Random Forrest Regressor

4

Upload



02 Service - 4 Model Explanation

Dashboard

User can upload pickle model and data files to get a full global and local explanation dashboard to explain to audience.

Steps

1. Select the local data (.csv) file.
2. Select the explanation technique.
3. Enter feature names if applicable.
4. Press Upload button.

Learn More

1

1) Chose model file:
Choose Files rf_Model.pkl

1) Chose x test file:
Choose Files X_rf.pkl

1) Chose y test file:
Choose Files y_rf.pkl

2

Explanation Type
Dashboard

3

Model Type
Random Forrest Regressor

4

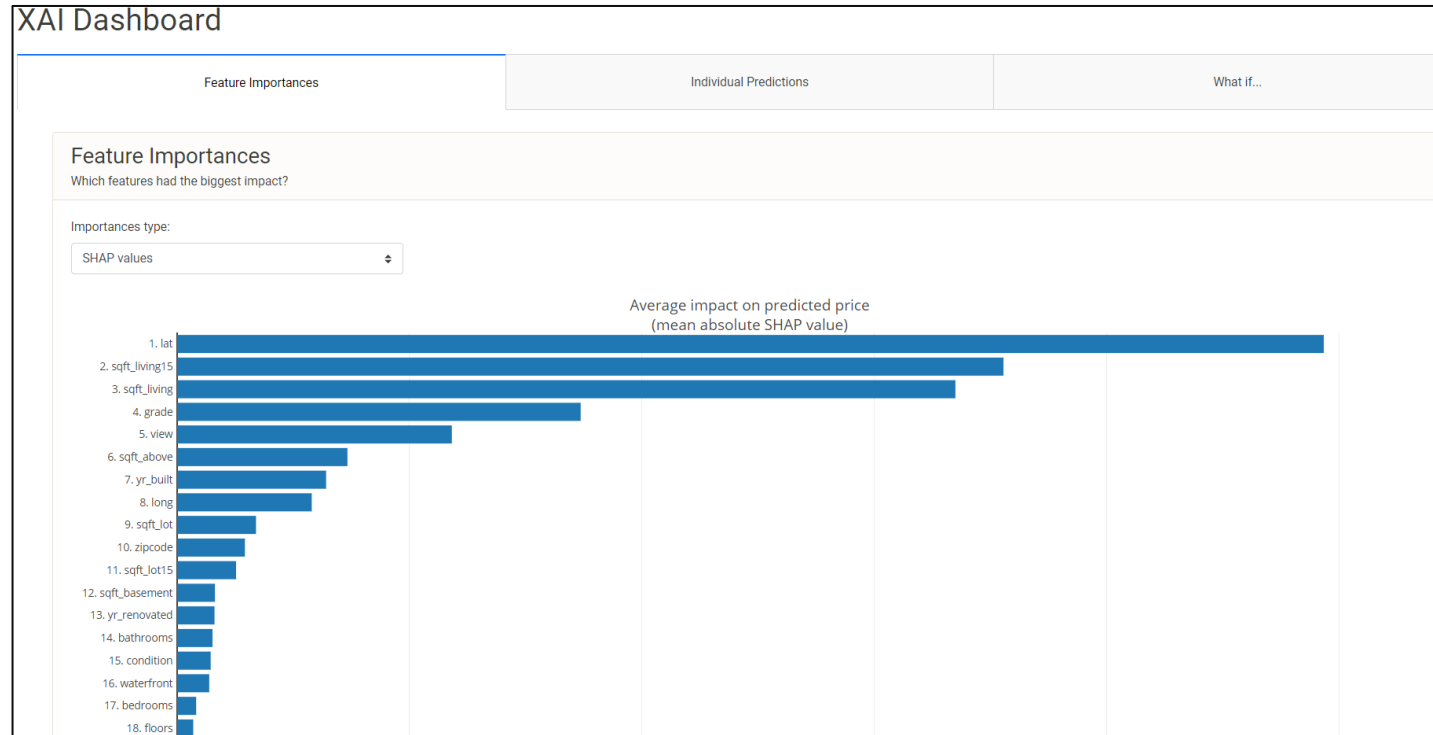
Upload



02 Service - 4 Model Explanation

Dashboard

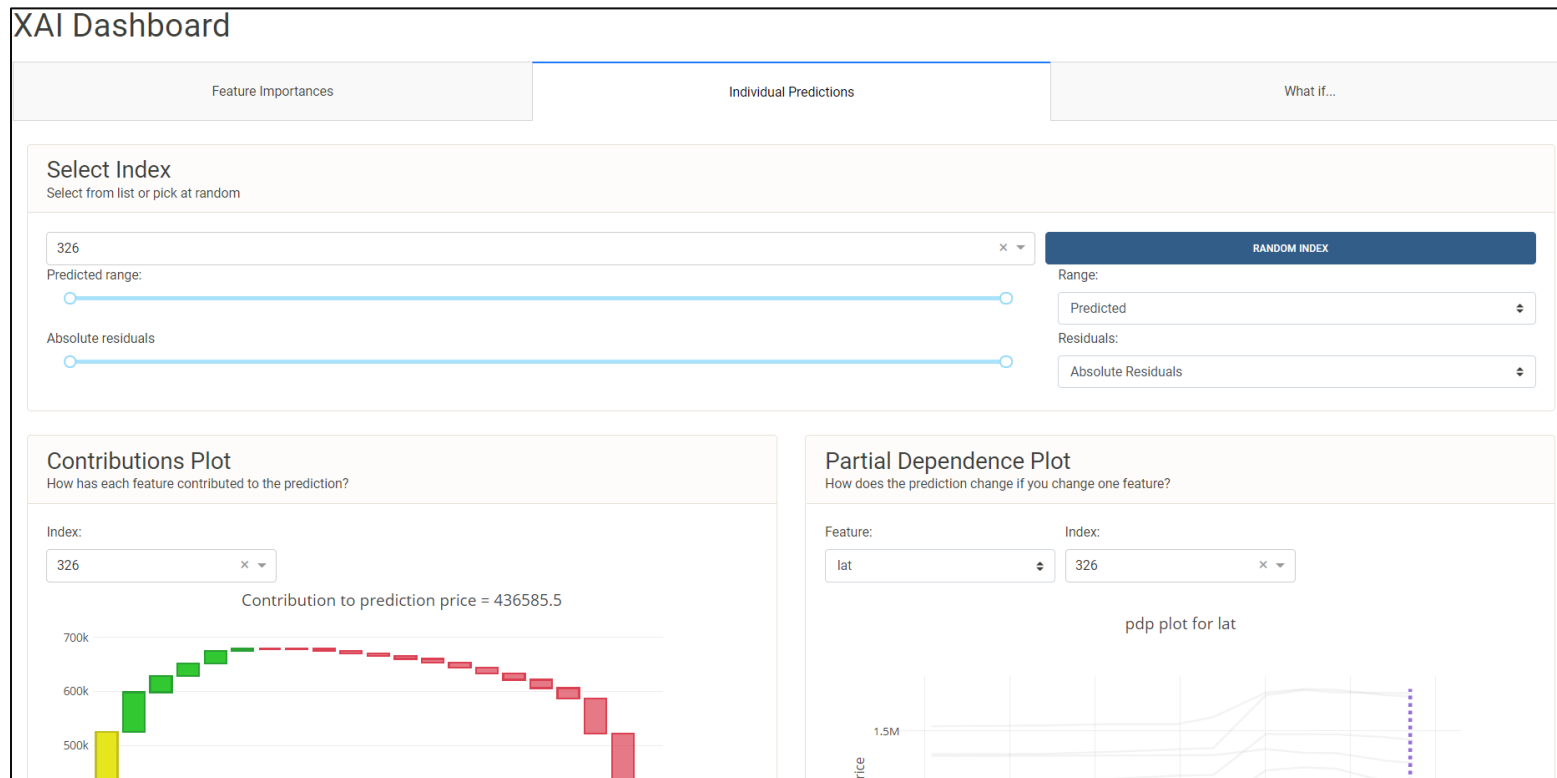
Feature Importance tab – SHAP values / Permutation Importance



02 Service - 4 Model Explanation

Dashboard

Individual Prediction – Select any instance to get SHAP explanation and PDP plot



02 Service - 4 Model Explanation

Dashboard

What-if Tab – create custom instance and get SHAP explanation and PDP plot

XAI Dashboard

Feature Importances

Individual Predictions

What if...

Feature Input

Adjust the feature values to change the prediction

lat	sqft_above	sqft_lot15	waterfront
47.6885	950	4050	0
sqft_living15	yr_built	sqft_basement	bedrooms
1320	1941	0	2
sqft_living	long	yr_renovated	floors
950	-122.337	0	1
grade	sqft_lot	bathrooms	
7	3960	1	
view	zipcode	condition	
0	98103	3	

Contributions Plot

How has each feature contributed to the prediction?

Contribution to prediction price = 453280.0

Partial Dependence Plot

How does the prediction change if you change one feature?

Feature:

Thank You